Solution for Chapter 2

Zimo Guo

2.1

(a) From standard PAC to 2-oracle PAC: D in standard PAC is any distribution. So for D_+ and D_-, P is less than or equal to ϵ. It's PAC-learnable in 2-oracle PAC model obviously.

(b) From 2-oracle PAC to standard PAC: The probability of standard PAC is

$$\begin{aligned} P[y \neq h(x)] &= P[(y = 1 \wedge h(x) = 0) \vee (y = 0 \wedge h(x) = 1)] \\ &\leq P[(y = 1 \wedge h(x) = 0)] + P[(y = 0 \wedge h(x) = 1)] \\ &\leq 2\epsilon \end{aligned}$$

ϵ is any positive number, so $P[y \neq h(x)] \leq \epsilon$.

2.4

This solution doesn't work. For non-concentric cycle, the h_s we get by training sample S may has a translation from the ideal hypothesis c_0. If the area of h_s is equal to $\pi r^2 - \epsilon$, that is, its area is equal to the small cycle in figure 2.5(b). In the case of figure 2.5(b), h_s move to the right top direction of c_0. Now three regions both have points locate in h_s. According to Gertrude's solution, the error of h_s should less than ϵ. But it's impossible obviously by the figure 2.5(b).

2.6

(a) Using the same notation in example 2.4, we still set four regions in the rectangle. Now some point will filp, while the sum of the area of the error should still be ϵ. Some points in h_s will contribute, so every region should have $(\epsilon - \epsilon_0)/4$, where ϵ_0 is the error of flipped points. Number of flipped points is less than m, so $\epsilon_0 \leq \epsilon\eta$.

Use the same process in example 2.4,

$$\begin{aligned} P[R(R_0^{'}) > \epsilon] &\leq 4(1 - \frac{\epsilon - \epsilon_0}{4})^m \\ &= 4(1 - \frac{\epsilon(1-\eta)}{4})^m \\ &\leq 4\exp\{-\frac{m\epsilon(1-\eta)}{4}\} \\ &\leq 4\exp\{-\frac{m\epsilon(1-\eta^{'})}{4}\} \end{aligned}$$

Let δ greater than or equal to the right side, get

$$m \geq \frac{4}{\epsilon(1-\eta^{'})} \log \frac{4}{\delta}$$

2.7

(a) h^* can give all correct predictions, so all $h^*(x_i)$ is equal to labels that randomly filpping has happend. Compare these with labels received by learners, all differences are caused by flipping. So the probability that the label of a training point received by the learner disagrees with the one given by h^* is equal to the probability that randomly filpping happen. That is $d(h^*) = \eta$.

(b) $d(h)$ could be contributed by two possible cases, one is filpping and correct prediction, another is no filpping and wrong prediction. The probability of filpping is η, correct prediction is $1-R(h)$. The probability of no filpping is $1-\eta$, wrong prediction is $R(h)$. Compute probability of two parts and sum them then get the target.

(c)

$$\begin{aligned} d(h) - d(h^*) &= \eta + (1-2\eta)R(h) - \eta \\ &= (1-2\eta)R(h) \\ &\geq (1-2\eta)\epsilon \\ &\geq \epsilon(1-2\eta^{'}) \\ &= \epsilon^{'} \end{aligned}$$

(d) Let $Z = 1_{y \neq h^*(x)}$, then $\hat{d}(h^*) = (1/m)\sum_{i=1}^{m} 1_{y_i \neq h^*(x_i)}$. $\hat{d}$ denotes the fraction and d denotes the probability, so $d(h^*) = E[\hat{d}(h^*)]$. Apply Hoeffding's inequality,

$$\begin{aligned} P[\hat{d}(h^*) - d(h^*) \geq t] &= P[(1/m)\sum Z_i - E[\hat{d}(h^*)] \geq t] \\ &= P[\sum Z_i - mE[\hat{d}(h^*)] \geq mt] \\ &\leq \exp\{-2mt^2\} \end{aligned}$$

Let $t = \frac{\epsilon^{'}}{2}$,

$$\begin{aligned} P &\leq \exp\{-\frac{m\epsilon^{'2}}{2}\} \\ &\leq \frac{\delta}{2} \end{aligned}$$

So

$$P[\hat{d}(h^*) - d(h^*) \geq \frac{\epsilon^{'}}{2}] \leq \frac{\delta}{2}$$

, that is

$$P[\hat{d}(h^*) - d(h^*) \leq \frac{\epsilon^{'}}{2}] \geq 1 - \frac{\delta}{2}$$

(e) Similar to (4), we want to prove

$$P[E[\hat{d}(h)] - \frac{1}{m}\sum Z_i \geq \frac{\epsilon^{'}}{2}] \leq \frac{\delta}{2}$$

According to Hoeffding's inequality,

$$\begin{aligned} leftside &= P[\sum Z_i - mE[\hat{d}(h)] \leq \frac{-m\epsilon^{'}}{2}] \\ &\leq \exp\{-\frac{m\epsilon^{'2}}{2}\} \\ &\leq \frac{1}{|\mathcal{H}|}\frac{\delta}{2} \\ &\leq \frac{\delta}{2} \end{aligned}$$

(f) Use the inequality in (3)(4)(5),

$$\begin{aligned} \hat{d}(h) - \hat{d}(h^*) &= [\hat{d}(h) - d(h)] + [d(h) - d(h^*)] + [d(h^*) - \hat{d}(h^*)] \\ &\geq -\frac{\epsilon^{'}}{2} + \epsilon^{'} - \frac{\epsilon^{'}}{2} \\ &= 0 \end{aligned}$$

So, when $m \geq \frac{2}{\epsilon^{'}} \log \frac{2}{\delta}$, with the probability at least $1 - \frac{\delta}{2}$, $\hat{d}(h) - \hat{d}(h^*) \geq 0$ holds.
Because $\frac{2}{\epsilon^{'}} \log \frac{2}{\delta} \leq \frac{2}{\epsilon^{'}}(\log |\mathcal{H}| + \log \frac{2}{\delta})$, and $1 - \frac{\delta}{2} > 1 - \delta$, following holds,
When $m \geq \frac{2}{\epsilon^{'}}(\log |\mathcal{H}| + \log \frac{2}{\delta})$, with the probability at least $1 - \delta$, $\hat{d}(h) - \hat{d}(h^*) \geq 0$.

2.12

According to Hoeffding's inequality, $P[\hat{R}_s(h) - R(h) \geq \epsilon] \leq \exp\{-2m\epsilon^2\}$. Let $\delta^{'} = p(h)\delta \geq$ the right side of Hoeffding's inequality. Then we get $\epsilon \geq \sqrt{\frac{\log \frac{1}{p(h)} + \log \frac{1}{\delta}}{2m}}$ with probability at least $1 - \delta^{'}$. $1 - \delta \leq 1 - \delta^{'}$, so the objective inequality is true.